

**Table 3**

Summary of relative uncertainties for linear correlations between  $C_h$  and selected predictive variables. Last 2 columns are  $p$  ratios of unfiltered parameter to filtered parameter, and each parameter to overall best parameter,  $A_r$ , respectively.

| Fluid/work    | Fig. 4 panels | Correlating variable        | MAE    | $R^2$  | $\frac{p_{un}}{p_{fit}}$ | $\frac{p}{p_{A_r}}$ |
|---------------|---------------|-----------------------------|--------|--------|--------------------------|---------------------|
| FC-72         | (a)           | $A_r$                       | 0.148  | 0.563  | 38.8                     | 1                   |
|               | (b)*          | $A_{r, \text{filt}}^*$      | 0.135* | 0.616* |                          |                     |
|               | (c)           | $R_d$                       | 0.177  | 0.413  | $1.98 \times 10^5$       | 0.730               |
|               | (d)           | $R_{d, \text{filt}}$        | 0.129  | 0.621  |                          |                     |
| Present       | (e)           | $\phi_m$                    | 0.146  | 0.541  | 33.7                     | 4.29                |
|               | (f)           | $\phi_{m, \text{filt}}$     | 0.130  | 0.596  |                          |                     |
|               | (g)           | $\kappa_{max}$              | 0.216  | 0.101  | $9.06 \times 10^8$       | 36.6                |
|               | (h)           | $\kappa_{max, \text{filt}}$ | 0.133  | 0.563  |                          |                     |
| FC-77<br>[17] | (a)           | $A_r$                       | 0.044  | 0.971  | 2.41                     | 1                   |
|               | (b)*          | $A_{r, \text{filt}}^*$      | 0.032* | 0.984* |                          |                     |
|               | (c)           | $R_d$                       | 0.138  | 0.781  | 13.5                     | 3.82                |
|               | (d)           | $R_{d, \text{filt}}$        | 0.045  | 0.960  |                          |                     |
|               | (e)           | $\phi_m$                    | 0.066  | 0.934  | 0.545                    | 14.9                |
|               | (f)           | $\phi_{m, \text{filt}}$     | 0.082  | 0.902  |                          |                     |
|               | (g)           | $\kappa_{max}$              | 0.268  | 0.173  | 12.6                     | 42.9                |
|               | (h)           | $\kappa_{max, \text{filt}}$ | 0.124  | 0.806  |                          |                     |
| Water<br>[17] | (a)*          | $A_r^*$                     | 0.080* | 0.795* | 1.08                     | 1                   |
|               | (b)           | $A_{r, \text{filt}}$        | 0.078  | 0.804  |                          |                     |
|               | (c)           | $R_d$                       | 0.099  | 0.735  | 1.09                     | 1.50                |
|               | (d)           | $R_{d, \text{filt}}$        | 0.085  | 0.749  |                          |                     |
|               | (e)           | $\phi_m$                    | 0.082  | 0.731  | 0.847                    | 1.54                |
|               | (f)           | $\phi_{m, \text{filt}}$     | 0.085  | 0.701  |                          |                     |
|               | (g)           | $\kappa_{max}$              | 0.142  | 0.127  | 4.90                     | 13.2                |
|               | (h)           | $\kappa_{max, \text{filt}}$ | 0.098  | 0.622  |                          |                     |

“un” – Unfiltered parameter.

“filt” – Filtered parameter.

\* Appropriately filtered  $A_r$  value provides the most certain correlation over all data and is used as the final basis of comparison.

to show somewhat consistent trends. Fig. 7 shows the approximate power relationships between  $t_{d,m}$  and  $q$  and  $\tau_m$  and  $q$  (Fig. 7a) and  $t_{d,m}/D_{d,m}$  and  $q$  and  $\tau_m/D_{d,m}$  and  $q$  (Fig. 7b). The results of the correlations are shown in Fig. 7c–f. The correlations evaluated are fully predictive in that they accept known or controllable parameters as independent variables. Those including the departure diameter – Fig. 7d and f – are based on values of  $D_d$  predicted by the present best-fit correlation with  $q$  and  $R_{d, \text{filt}}$  as expressed in Table 4 row (a). MAE values are lower for the correlations including  $D_d$ . The  $p$  ratio for correlating  $t_d$  with  $q$  and predicted  $D_d$  [panel (d)] versus with  $q$  alone [panel (c)] is about 2, but this is not significant considering the variance introduced by the additional factor. The  $p$  ratio for correlating  $\tau_d$  with  $q$  and predicted  $D_d$  [letter (e)] versus with  $q$  alone is less than 0.25, and is therefore a much less uncertain correlation form. Correlations between  $t_d$  or  $\tau_d$  and  $D_d$  take the form of a representative velocity and therefore agree qualitatively with the cited literature [52–54].

#### 4.4.2. Bubble growth curves

Comparison and discussion of experimental single-bubble growth rates from a number of studies was provided by Lee et al. [42]. They developed a semi-analytical correlation including both the inertially controlled growth and thermally controlled growth regimes and fit the correlation to a large set of experimental data. They found that, in the thermally limiting case (which from their analysis makes up the final 95% of the growth period), the extant data for a variety of fluids could be correlated well in the form  $D^* \sim (t^*)^a$ , where the asterisks indicate normalized quantities and the value of the exponent  $a$  could vary between 1/3 and 1/5. A fundamental issue with the extant data is that it is based on single isolated bubbles (characteristic of low heat fluxes or very smooth surfaces) and/or very small data sets.

In the present work bubble growth curves for all test conditions and surfaces followed a similar relationship between diameter and time when each bubble's growth curve was normalized by the